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COS700 Research Proposal

Diversity-Enhanced Evolutionary Optimisation of 3D Agents in the Framsticks Environment

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Abstract

Evolutionary algorithms have long been applied to the optimisation of simulated agents, yet finding agents whose centre of gravity (COG) follows a prescribed path across diverse and unknown physical environments remains an open challenge. This proposal describes a group research project that develops and evaluates Diversity-Enhancing NSGA-II (DE-NSGA-II) and a novel Adaptive DE-NSGA-II extension to optimise 3D agent morphologies and controllers within the Framsticks simulation platform, targeting the GECCO 2026 Automated Design Competition. The project investigates which combination of genetic encoding (f0 and f1), dissimilarity measure, and crowding strategy best enables generalisation across varied environmental configurations defined by parameters such as gravity, water level, terrain type, and initial agent orientation. Experiments evaluate the impact of gene-based and graph-based dissimilarity as crowding distance on solution quality and population diversity. The final system is benchmarked against the competition fitness metric, which measures how closely the evolved agent's COG trajectory matches a desired path.

Keywords: Evolutionary Algorithms, Framsticks, 3D Agent Design, Centre of Gravity Optimisation, DEAP, Genetic Encodings, GECCO Competition, NSGA-II, DE-NSGA-II, Quality-Diversity, Crowding Distance

1 Introduction and Motivation

Evolutionary design (ED) uses evolutionary algorithms (EAs) to automatically generate and optimise complex structures. One particularly challenging domain is the design of 3D simulated agents (creatures) whose motion follows a desired trajectory — a task that requires simultaneously optimising both body morphology and neural control systems. This project participates in the GECCO 2026 Automated 3D Design Competition, hosted within the Framsticks simulation environment, which challenges participants to evolve agents whose centre of gravity (COG) follows a desired path (e.g., a straight line in 3D, a sinusoidal oscillation, or jumping) across 10 diverse world configurations.

A critical challenge in this domain is premature convergence: fitness landscapes in evolutionary design are highly rugged and multimodal due to indirect genotype-to-phenotype mappings, and purely fitness-driven search tends to get trapped in local optima. This motivates the use of quality-diversity (QD) approaches that simultaneously optimise performance and population diversity. Building on prior work by Komosinski and Mensfelt (GECCO 2025), this project develops and evaluates Diversity-Enhancing NSGA-II (DE-NSGA-II) and a novel adaptive extension within the competition setting.

Evolutionary algorithms, including multi-objective approaches such as NSGA-II, have been widely used to optimise agent design and behaviour (Deb et al., 2002). This project therefore investigates how these algorithms can be designed and tuned to produce agents whose COG follows a desired spatial trajectory across varying environments. The core research question driving this project is: Can diversity-preserving, quality-diversity variants of NSGA-II — specifically DE-NSGA-II — outperform standard evolutionary baselines when optimising active 3D agents for centre-of-gravity trajectory tasks in the Framsticks environment?

2 Background

2.1 The Framsticks Environment

Framsticks is an open-source simulation platform for the evolution of 3D morphological structures controlled by recurrent neural networks (Komosinski and Ulatowski, 2009). Bodies are composed of mass-points connected by joints, and control systems consist of artificial neurons — sensors, effectors, and sigmoids — that can be freely interconnected. Two primary genetic encodings are relevant to this competition:

- f0 (low-level): A direct serialisation of the phenotype. Genetic operators work at the phenotypic level — mutation shifts part positions, rotates parts, and modifies neural connections; crossover splits the structure using a random plane in 3D space.
- f1 (high-level): A compact symbolic string encoding where individual characters map to body parts, branches, and neural connections. Mutation adds or removes symbols; crossover exchanges substrings between parents.

Both encodings are supported by the competition and produce agents that can be evaluated for arbitrary fitness criteria defined by the organisers.

2.2 The Competition Task

Submitted algorithms will be evaluated across 10 settings (environment x fitness function combinations). Each setting uses a maximised fitness function that measures similarity between the agent's COG trajectory and a desired path. Example fitness functions include 1.0/RMSE of 3D COG coordinates versus a desired line, or 1.0/sum-of-absolute-differences for sinusoidal bouncing of the form 0.5 + 0.1*sin(t/100.0).

Each run is budget-constrained to 100,000 fitness evaluations or 1 hour of algorithm runtime. Each setting is repeated 20 times; the 20 best-fitness values are averaged and normalised across all submissions to produce a final score.

2.3 Quality-Diversity Algorithms

Quality-diversity (QD) algorithms seek a diverse set of high-performing solutions rather than a single optimum (Chatzilygeroudis et al., 2021). Rather than converging to one peak, they maintain a population that covers the morphological or behavioural space broadly. QD methods — including MAP-Elites and Novelty Search with Local Competition (NSLC) — have demonstrated significant advantages in robotics and morphological evolution by avoiding premature convergence.

NSGA-II, a well-established multi-objective EA, can be adapted as a QD algorithm by treating both fitness and a diversity score as separate optimisation objectives. Its crowding distance mechanism, which normally estimates density in objective space, plays a critical role in maintaining population diversity.

2.4 DE-NSGA-II: The Key Inspiration

Komosinski and Mensfelt (GECCO 2025) introduced Diversity-Enhancing NSGA-II (DE-NSGA-II), a modification of NSGA-II that replaces the standard objective-space crowding distance with a dissimilarity-based crowding distance computed from the average pairwise dissimilarity between individuals in the population. Their paper evaluated this approach on Framsticks using two dissimilarity measures:

- Genetic dissimilarity (gene): Levenshtein distance computed on genotype strings — computationally efficient and encoding-agnostic.
- Phenotypic dissimilarity (graph): A graph-matching measure comparing 3D structural graphs of evolved bodies, accounting for vertex degree, neuron counts, geometric positions, and unmatched parts.

The key algorithmic change is that instead of sorting solutions by objective values to assign crowding distances, a pairwise dissimilarity matrix is computed and each individual's crowding distance is set to its average dissimilarity to all others in the population: I_i.crowding_dist = (1 / (|I| - 1)) * sum_{j != i} D_ij.

Finding	Implication for This Project
Standard NSGA-II achieved higher raw fitness; DE-NSGA-II significantly enhanced diversity in most conditions.	NSGA-II serves as the performance baseline; DE-NSGA-II is the diversity-focused challenger.
For f1 encoding on velocity tasks, graph-based crowding with gene as diversity score was Pareto-optimal.	f1 with graph crowding is the primary variant to test on active COG trajectory tasks.
Levenshtein (gene) measure was computationally efficient and broadly effective as a diversity score.	Gene measure is preferred where evaluation budget is tight.
A trade-off between diversity and fitness is representation- and task-dependent.	Motivates the Adaptive DE-NSGA-II variant that implements a continuous sigmoidal decay function on diversity.

3 Research Problem and Objectives

3.1 Problem Statement

How effectively can evolutionary algorithms be designed and tuned to produce 3D agents in the Framsticks environment whose centre of gravity follows a desired spatial trajectory across a range of unknown physical environments?

This overarching question decomposes into the following sub-problems that must be addressed:

1. Which genotype encoding — among those supported by Framsticks — allows evolutionary search to find high-quality COG-path solutions most efficiently?
2. Which evolutionary strategy (NSGA-II, DE-NSGA-II Gene, DE-NSGA-II Graph, or Adaptive DE-NSGA-II) best balances exploration and exploitation on this problem?
3. How should the fitness function and constraint-handling be formulated to reward COG-path similarity while respecting the structural constraints imposed by each evaluation environment?
4. Can replacing fitness-based crowding distance with dissimilarity-based crowding distance improve solution quality and diversity under the competition's 100,000-evaluation budget?
5. How well do evolved agents generalise to environment configurations not seen during the optimisation phase?
6. How can the population be successfully initialized to bypass the locomotion bootstrap problem without introducing human design bias?
7. How can a multi-phased adaptive algorithm transition smoothly from structural diversity to performance exploitation without inducing a destructive population selection shock?

Answering these questions requires both a solid understanding of the Framsticks simulation model and evolutionary computation theory. The scope of the research is bound to the problem settings, genotype encodings, and Python interface provided by the GECCO 2026 Framsticks competition.

3.2 Research Objectives

1. Implement DE-NSGA-II for the Framsticks competition environment using both f0 and f1 encodings with the FramsticksLibCompetition interface.
2. Compare DE-NSGA-II variants (gene-crowding, graph-crowding) against the standard NSGA-II baseline across competition-style settings.
3. Investigate adaptive crowding distance selection — dynamically decaying the structural diversity weight using a continuous sigmoidal cooling schedule mid-run to shift seamlessly from early exploration to late fitness convergence.
4. Evaluate the effect of encoding choice (f0 vs f1) on algorithm performance, integrating computational safeguards such as bounded graph-matching to protect the 1-hour competition execution budget.

4 Literature Study

4.1 Adaptive Intelligence through Evolutionary Optimization

The increasing complexity of modern autonomous systems, engineering design tasks, and large-scale optimization problems has positioned Evolutionary Multiobjective Optimization (EMO) as one of the most influential paradigms within computational intelligence. Unlike single-objective optimization, where the search process is directed toward a singular optimum, multiobjective optimization problems (MOPs) require the simultaneous optimization of multiple competing objectives, such as performance, stability, energy consumption, reliability, structural efficiency, and adaptability. In practical applications, these objectives are rarely independent. Instead, improving one objective frequently degrades another, creating a set of trade-off solutions commonly represented through the Pareto front (Fonseca & Fleming, 1998).

As optimization problems become increasingly realistic, the search process is further complicated by the introduction of constraints, high-dimensional decision spaces, expensive simulations, indirect genotype-to-phenotype mappings, and dynamic environmental interactions (Zhang et al., 2015). These characteristics collectively generate highly rugged search landscapes in which conventional optimization strategies struggle to maintain convergence, diversity, and computational efficiency simultaneously. Traditional static search frameworks often allocate computational resources inefficiently, leading either to premature convergence around locally optimal regions or excessive exploration within infeasible search spaces.

The field of Evolutionary Robotics (ER) exemplifies these difficulties particularly well. The evolutionary design of three-dimensional (3D) agents requires the simultaneous optimization of morphology and control systems within physically simulated environments. Such systems must navigate difficult terrain, maintain stability, adapt locomotion strategies, and coordinate mechanical interactions under uncertain environmental conditions. The coupling between morphology and behaviour introduces an additional layer of complexity because the effectiveness of a controller is intrinsically linked to the structure of the body it governs. Consequently, evolutionary search within 3D robotics must simultaneously address objective optimization, constraint satisfaction, morphological diversity, computational cost, and environmental adaptability.

Recent literature demonstrates a significant paradigm shift away from rigid, general-purpose optimization frameworks toward adaptive, stage-aware, and resource-autonomous architectures capable of dynamically modifying their behaviour according to the state of the

search process. Earlier generations of evolutionary algorithms relied heavily on static selection pressures, fixed cooperation strategies, and generalized diversity mechanisms that were often insufficient for highly constrained or high-dimensional optimization landscapes. Contemporary research increasingly argues that optimization systems must behave more intelligently by reallocating computational resources, adapting constraint pressure, maintaining structural diversity, and modifying exploration strategies in response to changing landscape conditions (Zuo et al., 2025; Horaguchi & Nakata, 2025).

This transition is particularly visible in three major research trajectories. The first concerns coevolutionary and stage-switching frameworks for constrained multiobjective optimization problems, where algorithms attempt to balance objective optimization with constraint satisfaction through adaptive subpopulation management. The second involves quality-diversity approaches that prioritize phenotypic novelty and behavioural diversity in order to prevent premature convergence within rugged evolutionary design spaces. The third concerns surrogate-assisted and multifactorial optimization frameworks developed for expensive high-dimensional problems, where computational efficiency becomes equally as important as convergence quality (Horaguchi & Nakata, 2025).

A fourth and increasingly prominent research trajectory concerns dynamic environments, where both objective functions and constraints may change over time. In dynamic multiobjective optimization problems (DMOPs), algorithms must not only converge toward an optimal trade-off surface but also track the movement of that surface as environmental conditions evolve. This requirement introduces unique challenges for diversity maintenance, convergence speed, and change detection that are distinct from those encountered in static optimization contexts (Jiang et al., 2022).Simultaneously, evolutionary robotics research has increasingly converged with these developments in optimization theory. The optimization of 3D robotic agents no longer focuses solely on controller evolution, but instead addresses the simultaneous co-evolution of morphology, behaviour, structural adaptability, and environmental interaction. Within rugged environments, robotic systems must continuously balance flexibility, stability, propulsion efficiency, and computational feasibility. Consequently, optimization algorithms designed for these systems must also become more adaptive, diversity-aware, and resource-conscious.

4.2. Major Algorithm Families in Evolutionary Multiobjective Optimization

4.2.1 Dominance-Based Algorithms

Dominance-based evolutionary algorithms constitute one of the foundational paradigms within multiobjective optimization research. Among these, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) remains the most influential and widely adopted framework due to its balance between computational simplicity, convergence capability, and diversity preservation (Ishibuchi et al., 2025; Komosinski & Mensfelt, 2025). Introduced by Deb et al. (2002), NSGA-II employs a fast non-dominated sorting mechanism alongside elitist selection and crowding-distance calculations to preserve a diverse approximation of the Pareto front.

The central mechanism underlying NSGA-II is the constrained dominance principle (CDP), which establishes a hierarchical relationship between feasible and infeasible solutions. Feasible solutions are always preferred over infeasible alternatives, while infeasible individuals are ranked according to their degree of constraint violation. This strategy provides a straightforward yet effective method for balancing objective optimization with feasibility preservation. Consequently, NSGA-II has become the standard baseline within constrained optimization research because of its transparency, reproducibility, and theoretical robustness (Fonseca & Fleming, 1998; Zuo et al., 2025; Ishibuchi et al., 2025).

The NSGA-II framework has also been extended in numerous directions beyond its original formulation: dynamic variants have been developed for non-stationary environments (Jiang et al., 2022), its crowding-distance mechanism has been replaced by surrogate-based selection criteria (Zhou et al., 2025), and its dominance-based selection has been incorporated into multi-objective neural architecture search (Long et al., 2025) and crossover operator comparison studies within path planning domains (Bostelmann-Arp et al., 2025).Despite its strengths, the literature consistently identifies several limitations that emerge when NSGA-II is applied to increasingly complex optimization landscapes. In rugged search spaces characterized by disconnected feasible regions or highly deceptive local optima, objective-space crowding mechanisms frequently fail to preserve meaningful structural diversity. Populations may rapidly converge toward locally optimal regions, resulting in premature convergence and the repeated rediscovery of similar solution structures. This issue becomes particularly severe within evolutionary robotics and 3D design problems, where multiple locomotion strategies or morphological configurations may exist simultaneously (Komosinski & Mensfelt, 2025). Furthermore, NSGA-II was originally designed for moderate-dimensional objective spaces and therefore struggles in many-objective optimization contexts where the dominance relationship between individuals becomes increasingly weak (Ishibuchi et al., 2025). As the number of objectives grows, populations may lose sufficient selection pressure,

reducing convergence efficiency and impairing diversity maintenance. Research has further shown that as the number of objectives increases to three or more, the convergence capability of dominance-based algorithms degrades substantially, making it difficult for standard frameworks such as NSGA-II and NSGA-III to improve upon high-quality heuristic initializations in combinatorial domains (Gong et al., 2025).

These limitations have motivated the emergence of decomposition-based, coevolutionary, and quality-diversity frameworks that attempt to overcome the weaknesses of purely dominance-based search. Nevertheless, NSGA-II remains indispensable within contemporary EMO research. Its relative simplicity allows researchers to clearly evaluate whether proposed adaptive mechanisms genuinely improve performance or merely introduce unnecessary algorithmic complexity. Consequently, modern frameworks frequently position NSGA-II as the primary baseline against which diversity-enhancing, surrogate-assisted, and coevolutionary strategies are evaluated (Zuo et al., 2025; Komosinski & Mensfelt, 2025).

4.2.2 Decomposition-Based Algorithms

Decomposition-based approaches such as MOEA/D and NSGA-III reformulate a multiobjective problem into multiple scalarized subproblems, each of which can be optimized individually while maintaining cooperation across the broader population. NSGA-III was specifically designed for many-objective optimization and relies heavily on predefined reference vectors to guide search behaviour, replacing the crowding-distance mechanism of NSGA-II with a structured reference-point-based selection strategy (Ishibuchi et al., 2025; Yadav et al., 2025).

These approaches offer several important advantages over purely dominance-based methods. They provide improved scalability in high-dimensional objective spaces, where the dominance relation becomes increasingly unreliable as a selective pressure. They also tend to produce better distributional spread across irregular Pareto fronts by directly controlling the mapping between reference vectors and population members, while enabling more structured population management through the decomposition framework itself.

However, research indicates that NSGA-III is highly sensitive to Pareto front geometry. On regular triangular fronts, the algorithm behaves effectively as a true decomposition-based method, with reference vectors naturally aligning with the objective space. On inverted or highly irregular fronts, normalization instability causes deterioration in performance because reference vectors can no longer meaningfully guide selection (Ishibuchi et al., 2025). Performance comparison studies across combinatorial optimization problems have further shown that decomposition-based algorithms may struggle to achieve competitive

hypervolume performance relative to mathematical programming-based relaxation methods when applied to large-scale constrained knapsack problems, even when extended to several thousand generations (Gong et al., 2025). Consequently, decomposition-based algorithms may struggle in combinatorial and highly irregular optimization landscapes without additional mechanisms for adapting the reference vector distribution to the true geometry of the problem.

4.2.3 Coevolutionary Algorithms

Coevolutionary frameworks attempt to improve exploration by evolving multiple interacting populations simultaneously, each subject to different selective pressures or problem formulations. The Coevolutionary Constrained Multiobjective Optimization (CCMO) framework represents a major advancement in this direction. CCMO maintains two concurrent populations: a main population that operates on the original constrained problem, and a helper subpopulation that solves an unconstrained relaxation of the same problem. By allowing the helper population to explore regions that would be inaccessible under strict feasibility constraints, the framework facilitates traversal through infeasible barriers and improves the diversity of solutions provided to the main population (Zuo et al., 2025).

Several CCMO variants have been proposed to address specific resource-allocation challenges inherent to the dual-population architecture. In CCMO-N, the helper subpopulation is terminated once it has converged, thereby avoiding the waste of computational resources on redundant exploration after the feasible front has been identified. In CCMO-SO, the offspring generation capacity of the helper population is reduced, reallocating function evaluations toward the refinement of feasible regions. CCMO-SC employs strong cooperation mechanisms that combine individuals from both populations during offspring generation, enabling more direct knowledge transfer between constrained and unconstrained search processes.

Coevolutionary frameworks provide strong exploration capabilities, particularly in landscapes with complex constraint geometries (Zhang et al., 2015; Gan & Zheng, 2011). However, they introduce significant resource-allocation challenges because maintaining an unconstrained helper population indefinitely may become computationally wasteful once the feasible front has already been identified. This observation has motivated the development of adaptive stage-switching mechanisms capable of dynamically modifying cooperation intensity and population structure according to the current state of the search (Zuo et al., 2025). An analogous stage-switching principle has been applied to decision-space-constrained binary problems, where one stage focuses exclusively on reducing constraint violations until feasibility is achieved, after which a second stage transitions to hill-climbing over the objective space (Honjo Ide et al., 2025). This architecture highlights that effective constraint handling in

complex constrained domains may require not only structural separation between feasibility-focused and objective-focused subpopulations but also a principled mechanism for switching between them based on search-state feedback.

4.2.4 Quality-Diversity Algorithms

Quality-Diversity (QD) algorithms shift the objective of optimization away from discovering a single optimal solution toward identifying a diverse collection of high-performing solutions that span the space of possible behaviours or morphologies. This shift is motivated by the observation that traditional evolutionary algorithms frequently reward globally dominant morphologies, causing populations to repeatedly rediscover identical body plans across independent runs and failing to explore the full range of viable strategies (Komosinski & Mensfelt, 2025).

Representative QD approaches include Novelty Search with Local Competition (NSLC), MAP-Elites, and Diversity-Enhancing NSGA-II (DE-NSGA-II). These algorithms explicitly reward morphological novelty, behavioural diversity, structural dissimilarity, and niche formation, ensuring that the resulting population encompasses a broad repertoire of qualitatively distinct solutions rather than converging toward a narrow region of the fitness landscape. MAP-Elites in particular has demonstrated strong performance across a range of evolutionary robotics benchmarks by maintaining an illumination of the behaviour-performance space, enabling efficient recovery of high-quality solutions across many different behavioural niches simultaneously (Komosinski & Mensfelt, 2025).

This shift toward quality-diversity search is particularly important within 3D evolutionary design, where multiple viable locomotion strategies may exist simultaneously and the discovery of a single high-performing solution provides limited insight into the broader space of adaptive possibilities (Sims, 1994). By preserving structural and behavioural heterogeneity, QD algorithms increase the robustness of evolutionary search and reduce susceptibility to the premature convergence that frequently afflicts purely objective-driven frameworks (Komosinski & Mensfelt, 2025).

4.3 Evolutionary Robotics and the Design of 3D Agents

4.3.1 The Evolutionary Robotics Paradigm

Evolutionary Robotics (ER) applies evolutionary computation to the automated design of autonomous robotic systems. Rather than manually engineering robotic behaviours through explicit programming or hand-crafted control laws, ER evolves populations of candidate solutions through iterative processes of selection, mutation, and recombination. This enables the discovery of unconventional and unintuitive designs that may substantially surpass human-generated solutions, particularly in complex environments where the interaction between morphology and behaviour produces emergent capabilities that are difficult to anticipate through deliberate design.

A major transition within ER has been the movement from controller-only evolution toward the simultaneous co-evolution of morphology, neural control systems, sensor configurations, and behavioural strategies. This transition reflects a growing recognition within the field that intelligence is fundamentally embodied: behaviour does not emerge solely from the structure of a control system, but arises through the interaction between body structure, controller dynamics, and the physical environment in which the agent operates (Maeda, 1997). The morphology of a robotic agent constrains and shapes the repertoire of behaviours that its controller can express, while conversely, the requirements of the control task influence which morphological configurations prove evolutionarily viable. This deep interdependence between body and brain makes the joint optimization of morphology and control one of the most challenging and consequential problems within evolutionary computation (Sims, 1994; Komosinski & Mensfelt, 2025).

4.3.2 Fitness Functions in 3D Agent Design

Fitness functions define the selective pressures driving evolutionary adaptation and are therefore among the most consequential design decisions in any evolutionary robotics system (Chaudhuri et al., 2022). The choice of fitness function determines not only which solutions are rewarded, but also the structure of the search landscape that the evolutionary algorithm must navigate. In 3D agent design, fitness functions typically span several broad categories, each capturing a distinct dimension of agent performance (Komosinski & Mensfelt, 2025).

Movement metrics constitute the most commonly employed category, encompassing measures such as locomotion velocity, total distance travelled across evaluation episodes, terrain coverage breadth, and locomotion energy efficiency. Mechanical metrics capture structural and dynamical properties of the agent, including centre-of-gravity stability, torque production capacity, load-bearing capability, and resistance to structural failure under environmental stress. Goal-specific objectives extend fitness evaluation to task-dependent criteria such as climbing performance on inclined surfaces, grasping and manipulation competence, path-following accuracy, and jumping height or distance (Xiao et al., 2025;

Chaudhuri et al., 2022). Safety constraints impose penalties for undesirable behaviours including collision events, catastrophic falls, structural damage, and excessive energy expenditure.

More recently, novelty metrics have been incorporated into fitness function designs, rewarding morphological diversity, behavioural uniqueness, and structural innovation independently of performance outcomes. The integration of novelty metrics has become increasingly important within the literature because purely objective-driven optimization frequently converges toward structurally homogeneous populations in which all individuals share similar body plans despite potentially large differences in their underlying genetic encodings. By explicitly rewarding dissimilarity, novelty-augmented fitness functions encourage the exploration of qualitatively distinct regions of the morphological design space (Komosinski & Mensfelt, 2025).

4.4. Diversity Enhancement in Evolutionary Design

4.4.1 The Problem of Premature Convergence

Premature convergence is one of the most severe and well-documented challenges in evolutionary design. Traditional evolutionary algorithms frequently reward morphologies that exploit easy pathways toward fitness improvement, creating strong selection pressure toward a narrow set of body plans that may represent locally optimal rather than globally superior solutions. Over time, this tendency causes populations to cluster around a restricted range of morphological configurations, progressively reducing genetic and phenotypic diversity and eliminating the potential to discover qualitatively different strategies through subsequent mutation and recombination (Komosinski & Mensfelt, 2025).

Within rugged 3D environments, premature convergence carries particularly serious consequences. Once a population has collapsed around a narrow morphological archetype, the evolutionary algorithm loses the capacity to discover alternative locomotion strategies that may prove more effective in different terrain conditions, environmentally-specific adaptations that require qualitatively different body structures, structurally innovative morphologies that exploit unconventional physical principles, and robust behavioural repertoires capable of generalizing across a range of environmental challenges. Preventing premature convergence therefore requires explicit mechanisms for maintaining population diversity throughout the evolutionary process, rather than treating diversity as a secondary consequence of selection and recombination (Komosinski & Mensfelt, 2025).

4.4.2 Dissimilarity-Based Diversity Metrics

DE-NSGA-II addresses the limitations of objective-space crowding by replacing the standard crowding distance mechanism with dissimilarity-based diversity measures that operate directly on the genetic or phenotypic characteristics of individuals (Komosinski & Mensfelt, 2025). Rather than measuring how isolated a solution is within the objective space, dissimilarity-based metrics assess how structurally distinct it is from neighbouring individuals in the population, providing a more direct representation of morphological diversity.

Two major categories of dissimilarity are utilized within this framework. Genetic dissimilarity measures structural differences in underlying genetic encodings through metrics such as Levenshtein distance for sequence-based representations, tree-edit distance for hierarchically structured genomes, and genome graph distance for directed graph encodings (Komosinski & Mensfelt, 2025). Phenotypic dissimilarity instead measures observable differences between realized morphological structures through metrics such as morphological graph distance, limb configuration analysis, and behavioural trajectory divergence across evaluation episodes (Komosinski & Mensfelt, 2025). Phenotypic diversity is regarded as particularly valuable within the literature because genetically distinct solutions may still produce structurally similar morphologies through convergent developmental pathways, meaning genetic dissimilarity alone is insufficient to guarantee meaningful morphological heterogeneity within the population.

4.4.3 Quality-Diversity Search in Framsticks

The Framsticks simulation environment provides a particularly valuable testbed for evolutionary design research because of its flexible agent representation and physically realistic simulation dynamics (Komosinski & Mensfelt, 2025). Agents in Framsticks are constructed from elastic stick-based morphological components and controlled through neural systems containing both sensor and effector elements, enabling the evolution of a broad range of locomotion strategies and body configurations within a single unified framework.

Research employing DE-NSGA-II within the Framsticks environment has consistently demonstrated that dissimilarity-based diversity maintenance produces qualitatively superior outcomes compared to purely objective-driven approaches (Komosinski & Mensfelt, 2025). Specifically, dissimilarity-aware selection encourages niche formation by maintaining structurally distinct locomotion strategies within the population simultaneously, avoids domination by single morphology classes that would otherwise suppress exploration of alternative body plans, and enhances robustness by ensuring that the evolutionary search maintains a broader repertoire of solutions capable of performing effectively across multiple locomotion tasks. This body of work represents a meaningful transition from traditional single-objective optimization toward Quality-Diversity search, where the goal is not merely

discovering the single highest-performing solution but generating a diverse and adaptive population of effective designs (Komosinski & Mensfelt, 2025).

4.5 Dynamic Multiobjective Optimization

Evolutionary Dynamic Multiobjective Optimization (EDMO) addresses problems in which objective functions, constraints, or environmental parameters change over time (Jiang et al., 2022). Unlike static multiobjective optimization, where the goal is to converge toward a fixed Pareto front, EDMO requires algorithms to continuously track a moving Pareto set as the problem landscape evolves. Environmental changes may cause previously optimal solutions to become suboptimal or infeasible, requiring algorithms to balance the reuse of historical search knowledge against the need to adapt rapidly to new conditions. Maintaining population diversity simultaneously becomes critically important, because a diverse population is more likely to contain individuals near the new Pareto front after an environmental change (Jiang et al., 2022). Dynamic environments are classified by whether changes affect the Pareto Set, the Pareto Front, or both, and by the origin of dynamics—changes in decision variables, objective functions, time-dependent interactions, or time-varying constraints (Jiang et al., 2022). Change detection methods span random re-evaluation, sensor-based detection, and population-based statistical testing. Change response strategies include hypermutation, prediction-based methods using historical Pareto set trajectories, memory-based retrieval of past solutions, and robustness-based approaches that seek solutions tolerant to environmental variation (Jiang et al., 2022). These insights are directly applicable to evolutionary robotics, where agents evolved under fixed conditions may perform poorly when conditions change, and stage-aware diversity maintenance, prediction-based adaptation, and multi-population change response are therefore relevant to evolving morphologically robust and behaviourally flexible 3D agents (Jiang et al., 2022).

4.6. Weakly Constrained Multi-Robot Systems

4.6.1 Constraint Architectures in Multi-Robot Systems

Multi-robot systems (MRS) are increasingly deployed for terrain traversal and task execution in environments that are inaccessible or impractical for isolated robotic agents. By distributing

locomotion and manipulation tasks across multiple cooperating agents, MRS can achieve levels of terrain coverage, payload capacity, and environmental adaptability that far exceed those of equivalent single-robot systems (Maeda, 1997). The design of effective multi-robot systems therefore requires careful consideration of the architectural relationship between agents, including the degree and type of physical and communicative coupling that governs their cooperative behaviour.

Three major architectural categories define the landscape of multi-robot constraint design (Xiao et al., 2025). Virtual Constraint Systems (MRS-VC) enable coordination through communication without physical links between agents, offering high flexibility and simplicity of deployment with low mechanical coupling. However, these systems suffer from relatively weak collective propulsion and reduced stability when navigating steep or irregular terrain, because they lack the mechanical integration that would allow agents to directly share forces and support one another structurally. Strong Constraint Systems (MRS-SC) address these limitations by rigidly connecting modular robot units, thereby achieving high traversability, strong collective propulsion, and structural stability on challenging terrain. The cost of this integration is high computational complexity in motion planning, reduced flexibility for deployment in variable environments, and difficult motion-compliance coordination between rigidly coupled units.

Weakly constrained systems represent a third architectural category that attempts to balance the flexibility of virtual constraint systems with the structural capability of strongly constrained architectures. By allowing the degree of mechanical coupling between agents to be modulated dynamically according to environmental conditions, weakly constrained systems can achieve task-adaptive behaviour that neither purely communicative nor rigidly coupled architectures can match.

4.6.2 Controllable Weak Constraints and Adaptive Stiffness

The Multi-Robot System with Controllable Weak Constraints (MRS-CWC) represents a significant advancement in adaptive robotic architectures (Xiao et al., 2025). The system utilizes McKibben Pneumatic Artificial Muscles as actuating elements in its interconnection mechanisms, enabling selective partial constraint application and dynamic stiffness

modulation across the coupling structure (Xiao et al., 2025). The stiffness coefficient of the inter-robot connections adapts in response to environmental torque estimation: softening occurs when environmental resistance becomes excessive to avoid mechanical failure or locomotion inefficiency, while stiffening occurs when propulsion efficiency must be maximized to overcome environmental resistance.

This adaptive mechanism allows multi-robot systems to traverse rugged terrain more effectively than static alternatives by preserving manoeuvrability in complex environments, reducing computational overhead through simplified environmental modeling, and operating effectively without requiring detailed prior knowledge of environmental conditions. Experimental results have demonstrated superior navigation completion rates compared to both constantly soft systems, which lack sufficient propulsion to overcome difficult terrain, and rigid motorized-joint architectures, which lack the compliance required for safe navigation in unstructured environments (Xiao et al., 2025).

The concept of controllable stiffness also provides a compelling metaphor for adaptive search frameworks within EMO. Just as robotic systems dynamically regulate physical rigidity according to terrain conditions and resistance forces, future optimization algorithms may dynamically regulate constraint adherence according to search landscape resistance, tightening constraint enforcement when the population is comfortably within feasible regions and relaxing it when infeasible barriers impede convergence. This conceptual parallel between physical adaptivity and algorithmic adaptivity highlights the deep theoretical connections between adaptive robotics and adaptive optimization that motivate the integrated research agenda explored in this review.

4.7 Critical Analysis

4.7.1 From Static to Adaptive Architectures

A major pattern emerging across the literature is the gradual abandonment of rigid optimization architectures in favour of adaptive, state-aware systems capable of dynamically modifying

search behaviour throughout the evolutionary process. Earlier generations of evolutionary algorithms typically relied on fixed exploration and exploitation strategies, static constraint-handling mechanisms, and globally uniform selection pressures. While these approaches performed adequately on relatively simple optimization landscapes, they proved increasingly insufficient for constrained, high-dimensional, or morphology-dependent search problems where the relationship between algorithmic behaviour and landscape structure changes significantly as the search progresses (Zuo et al., 2025; Horaguchi & Nakata, 2025).

Modern research instead emphasizes the importance of computational autonomy. Rather than treating optimization as a static procedure with fixed algorithmic parameters, contemporary frameworks increasingly monitor the state of the search process itself and adjust their behaviour accordingly. Coevolutionary systems dynamically adjust subpopulation cooperation according to convergence and feasibility conditions (Zuo et al., 2025). Diversity-enhancing frameworks modify selection pressure to preserve structural novelty in regions where objective-space diversity alone is insufficient to prevent premature convergence (Komosinski & Mensfelt, 2025). Multifactorial optimization systems share information across related subproblems in order to maximize the utility of each function evaluation under severe computational constraints (Gan & Zheng, 2011; Horaguchi & Nakata, 2025).

This transition toward adaptive architectures reflects a broader conceptual shift within computational intelligence. Optimization is no longer viewed simply as the process of converging toward a Pareto-optimal region. Instead, modern evolutionary systems increasingly behave as adaptive search ecologies that regulate exploration, cooperation, diversity, and resource allocation according to environmental conditions and search-state feedback. In this sense, the development of stage-switching mechanisms, controllable stiffness models, and dynamic diversity maintenance strategies represents not merely a collection of isolated technical improvements, but a unified movement toward autonomous optimization systems capable of self-regulating their own search behaviour in response to the demands of the problem at hand.

4.7.2 Diversity as a Functional Necessity

Earlier EMO frameworks treated diversity primarily as a mechanism for improving distributional spread across the Pareto front, ensuring that the final solution set provided decision-makers with a comprehensive representation of trade-off alternatives. While this remains a valuable property, recent research has significantly expanded the conceptual role of diversity within

evolutionary optimization, reframing it as a functional necessity that serves multiple essential purposes simultaneously (Komosinski & Mensfelt, 2025).

Diversity is increasingly understood as a source of population-level robustness that allows algorithms to recover from unfavourable search trajectories and maintain exploration capability throughout the optimization process. It functions as a defence against premature convergence by ensuring that no single morphological or behavioral archetype dominates the population to the exclusion of all alternatives (Komosinski & Mensfelt, 2025). It provides the raw material for discovering alternative behavioural strategies that may prove superior in different environmental conditions or evaluation scenarios. And within the context of evolutionary robotics, diversity constitutes a prerequisite for adaptive embodied intelligence, because agents must possess qualitatively different physical and behavioral capabilities to navigate the full range of challenges posed by complex and unpredictable environments. Within 3D evolutionary design, phenotypic diversity becomes especially important because structurally distinct agents may solve environmental challenges using entirely different locomotion principles, and the loss of any major locomotion archetype from the population permanently forecloses that adaptive pathway (Komosinski & Mensfelt, 2025).

4.7.3 Reality Gap Limitations

The development of evolutionary systems capable of enabling lifelong morphological adaptation across sequential terrains or dynamically changing task requirements represents a significant frontier within evolutionary robotics, requiring new formulations of the fitness landscape, novel encoding strategies that support developmental plasticity, and optimization frameworks capable of balancing the preservation of previously acquired capabilities with the exploration of new adaptive strategies.

Simulation-evolved agents frequently exhibit severe performance degradation when transferred into physical environments, a phenomenon widely referred to as the reality gap. This gap arises from systematic discrepancies between simulated and physical conditions, including fabrication inconsistencies in physical implementations of evolved morphologies, inaccurate physics modeling that fails to capture the full complexity of material deformation, contact dynamics, and frictional interactions, numerical instabilities that produce qualitatively different trajectories in physical hardware compared to simulation, and energy dissipation effects that are difficult to model accurately without detailed material characterization. Bridging the reality gap requires either significantly more accurate simulation environments, explicit co-optimization of simulation fidelity and controller performance, or evolutionary strategies that favour robust solutions capable of maintaining performance despite environmental variability.

5 Proposed Methodology

The research methodology adopted for this project is Proof by Demonstration (Johnson, 2006). Proof by Demonstration involves constructing an artefact — in this case, the DE-NSGA-II algorithm variants described below — and iteratively refining it through repeated cycles of implementation, testing, and evaluation. Iterative refinement means that each variant is built, fine-tuned, and assessed in turn, with the results and any issues identified during the testing of one variant used to inform the design and configuration of the next, gradually progressing towards the novel Adaptive DE-NSGA-II contribution. Each algorithm variant will first be developed and evaluated individually under controlled conditions to demonstrate that it functions as intended and to assess its strengths and weaknesses. Thereafter, the performance of all variants will be compared to determine which combination of genetic encoding, dissimilarity measure, and crowding strategy best generalises across the competition's environments, thereby demonstrating the value of the diversity-preserving and adaptive contributions of this project. The remainder of this section describes the artefact to be built and the phases through which it will be iteratively developed, refined, and demonstrated.

5.1 Algorithm Design

The following algorithm variants will be implemented within a unified DEAP-based framework, each building on the previous to isolate the contribution of each change:

Variant	Description
Variant 1 — NSGA-II (Baseline)	Standard multi-objective NSGA-II treating COG-path fitness and a diversity score as dual objectives using objective-space crowding distance. Tested with both f0 and f1 encodings. Reproduces the baseline from Komosinski and Mensfelt (2025).
Variant 2 — DE-NSGA-II (Gene Crowding)	Crowding distance replaced with average pairwise Levenshtein distance on genotype strings. Computationally efficient and applicable to both encodings. Diversity objective: average gene dissimilarity to population.
Variant 3 — DE-NSGA-II (Graph Crowding)	Crowding distance replaced with average pairwise graph-matching dissimilarity on 3D phenotypic structures, accounting for vertex degree, neuron counts, and geometric positions. Optimization safeguard: Graph comparisons will use a structural depth bound and a memorized sub-graph cache to remain strictly within the 1-hour execution limit.
Variant 4 — DE-NSGA-II (Hybrid Crowding)	Crowding distance computed as a weighted combination of gene and graph dissimilarities, with the weight tuned per encoding type. Balances computational efficiency with structural diversity.
Variant 5 — Adaptive DE-NSGA-II (Novel Contribution)	*Adaptive Sigmoidal Selection: T*his variant implements a continuous sigmoidal decay function. It smoothly transitions selection pressure from dissimilarity-based crowding (promoting structural exploration early) to objective-space crowding (promoting exploitation late) to completely prevent population selection shock.

5.2 Implementation Details

Component	Detail
Language	Python (as required by the competition rules)
Framework	DEAP library, extended with DE-NSGA-II selection operators
Interface	FramsticksLibCompetition class (as specified by the competition)
Encodings	Both f0 and f1; encoding selected based on preliminary experiments per task type
Constraint handling	Solutions violating world configuration constraints are penalised or replaced as returned by the evaluator
Dependencies	deap, framspy (Python interface to Framsticks), python-Levenshtein for efficient edit distance computation

5.3 Experimental Design

Consistent with the iterative, demonstration-driven approach described above, the artefact will be developed and demonstrated across three phases:

Phase	Description
Phase 1 — Controlled internal experiments	Using publicly documented simulation parameters as proxies for competition settings, run all 5 variants x 2 encodings x 4 proxy tasks x 20 repeats. Measure: mean best fitness, average gene/graph diversity of best 20 solutions, convergence curves.
Phase 2 — Sensitivity analysis	Vary key parameters: population size (50-200), tournament size, the adaptive switching threshold τ, and the weight between gene/graph in Variant 4.
Phase 3 — Evaluation and testing	Select the best-performing variant and encoding per task type. Test in a clean environment using FramsticksLibCompetition.

5.4 Evaluation Metrics

Metric	Description
Mean Best Fitness	Average of best fitness across 20 runs per setting
Normalised Score	Competition normalisation across all submissions
Gene Diversity	Average Levenshtein distance of the 20 best genotypes per run
Graph Diversity	Average phenotypic graph dissimilarity of the 20 best structures per run
Convergence Rate	Number of evaluations to reach 90% of final best fitness
Generalisation Score	Average fitness on held-out .sim environments not seen during training

Statistical significance will be assessed using the Kruskal-Wallis test followed by Dunn's post-hoc tests ($\alpha = 0.05$), consistent with the methodology of Komosinski and Mensfelt (2025).

5.5 Constraint and Bootstrap Problem

To address the locomotion bootstrap problem identified in Section 3.3—where early, random populations fail to generate a baseline movement signal—the framework will implement a Minimal-Viable Morphological Filter. Before consuming the simulation budget, newly generated genotypes must pass a lightweight structural check requiring a non-zero count of interconnected muscles/joints and a balanced geometric layout. Non-viable "static blobs" are filtered and re-sampled at initialization. This ensures a functional fitness landscape exists from generation zero without introducing restrictive human-engineered design bias.

6 Expected Contributions

1. Empirical extension of DE-NSGA-II to active trajectory-following tasks, generalising the findings of Komosinski and Mensfelt (2025) beyond the vertical-position (passive) and velocity (active) tasks studied in that paper.
2. Novel Adaptive DE-NSGA-II algorithm that dynamically balances exploration and exploitation within the evaluation budget — a contribution not present in the original DE-NSGA-II paper.
3. Encoding-task interaction analysis: practical guidance on which dissimilarity measure and encoding combination is most effective for trajectory-optimisation tasks in Framsticks.

7 Work Plan and Team Responsibilities

Milestone	Week	Responsible
Literature review and baseline replication	Weeks 1-6	All members
f0/f1 encoding + gene/graph dissimilarity implementation	Weeks 3-8	All Members
DE-NSGA-II Variants 1-4 implementation	Weeks 7-12	All Members
Adaptive DE-NSGA-II (Variant 5)	Week 9-12	Member 1
Phase 1 experiments (internal settings)	Weeks 11-16	All Members
Parameter sensitivity analysis	Week 13-18	Member 1 & 3
Statistical analysis and visualisation	Week 17-20	Members 2 & 3
Clean environment testing + submission packaging	Week 19-22	All members
Honours report write-up	Weeks 19-22	All members

8 Planning

The project plan is shown in Figure 2. Each column represents 2 weeks of the 22-week project period from 1 June to 31 August. Green cells indicate the expected active period for each task.

Task	30/05	13/06	27/06	11/07	25/07	08/08	22/08	05/09	19/09	03/10	17/10	31/10
Literature Review	■	■	■									
Framsticks Setup & Familiarisation		■	■	■								
Implement Baseline NSGA-II (DEAP)			■	■	■							
Implement DE-NSGA-II Variants				■	■	■	■					
Implement Adaptive DE-NSGA-II					■	■	■					
Experiment: Encoding Comparison						■	■	■				
Experiment: Parameter Tuning							■	■	■			
Iterative Improvement & Testing								■	■	■	■	
Final Evaluation & Analysis									■	■	■	■
Write Final Research Paper										■	■	■

■ Active period for task

Figure 2: Gantt Chart of Project Plan (01 June – 31 August)

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